

The IT Industry's North Star



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One of the universal truths of being a leader in the semiconductor industry is that you *will* be asked some variant of the question, “What do you think about the future of Moore’s Law?”

I was an electrical engineering undergraduate student when Gordon Moore famously predicted that the number of transistors on a chip would double about every two years. For a generation of engineers, Moore’s Law was the goal against which they measured success. How many transistors could you cram onto a single integrated circuit?

Times Have Changed

For many years, Moore’s Law was the best barometer to describe computing technology’s potential. More transistors translated into more computing functionality. Over time, however, in its endless pursuit of Moore’s Law, the IT industry increased transistor counts and lowered die sizes to the detriment of the chip design itself. The industry kept adding transistors, but didn’t always design technology that was more useful or beneficial. The result was people paid for technology—in the form of more transistors—that they didn’t want and didn’t need.

Today, Moore’s Law illustrates the age-old lesson that bigger is not always better. Think of it in terms of a painter who has a bigger and bigger canvas to work with each time, but keeps painting the same picture.

Moore’s Law is still an important observation in terms of what it means for the economics of the semiconductor industry. The bigger canvas still has intrinsic value, especially when it doesn’t come with a cost premium. But to treat Moore’s Law as the guiding principle for semiconductor innovation is to be blind to the needs of today’s technology consumers.

What Really Matters

While still important, raw performance is no longer the only benefit consumers look for in our products. Affordability, choice, and ease of use now play an equal, if not a greater role, in shaping technology purchase decisions.

Simple though they may sound, these new customer demands require no small change from our industry’s historical approach to innovation. Where we once began by understanding how to fit more transistors on a chip, we must now start by understanding the evolving and complex needs of our consumers. Successfully addressing the needs

of people, not stretching the limits of technology, must become our industry’s metric for successful innovation.

When inventor Bob Metcalfe needed to convince the world to adopt his Ethernet standard, he coined a maxim of his own. Metcalfe’s Law states that the value of a network is proportional to the square of the number of users of the system. Metcalfe’s Law puts people ahead of technology, and forces us to realise that technology is only as powerful as it is accessible. The network by itself is worthless. Only by having users successfully access it does it begin to hold value.

While Metcalfe’s observation originated in telecommunications, it unquestionably holds important implications for the entire IT community, especially those of us in the semiconductor industry. What good is a 3 gigahertz computer sitting in classroom if the teacher is technology illiterate? How truly mobile is a laptop if it only has two hours of battery life or if its processor burns up 90 watts of power? How affordable are our servers if the cost in energy to run them begins to outstrip the cost of purchasing the servers themselves?

New Laws

Our world is changing, and we, the IT industry, should rejoice in the role we have played in shaping a more productive and connected world. But if we are not careful, we run the risk of developing technologies that are no longer relevant to the most important needs of our consumers. While Moore’s Law was an important metric for understanding one era of technology, it is clear consumers have moved into a different era. It is time we follow suit.

Metcalfe’s Law, while not a useful forecast tool for chip design, nonetheless provides a critical and much needed North Star for the semiconductor industry to follow. It will force us to measure our value not in the number of transistors we can fit on a chip, but instead by the number of people who successfully use our technology. And it will help us express our potential not by what our processors can do in benchmark tests, but by the opportunities we create for the students, teachers, parents, patients, and doctors who are the real barometer of our future success. ■

Hector Ruiz joined AMD in January 2000. He is passionate about the role of technology in education and empowering the underprivileged. He currently serves on the American President’s Council of Advisors for Science and Technology (PCAST).